

The Hull-White 1 Factor Convexity Adjustment and the Special Case when the Hull-White and Ho-Lee Models are Equivalent

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Abstract

In what follows we quote the Hull-White 1 factor and Ho-Lee model dynamics and their corresponding eurodollar futures convexity adjustment formulas. We then show that, in the special case where the Hull-White mean reversion parameter is zero, the adjustment under the Hull-White and Ho-Lee models is identical.

Models

The Ho-Lee model takes the following form

$$dr(t) = \theta(t)dt + \sigma dB(t) \tag{1}$$

and the corresponding Hull-White 1 factor model is as follows

$$dr(t) = (\theta(t) + ar(t))dt + \sigma dB(t) \tag{2}$$

where

$r(t)$ = instantaneous interest rate at time, t

a = speed of mean reversion, $0 \leq a \leq 1$

$b(t)$ = long term mean reversion level

σ = volatility

$B(t)$ is a Brownian process at time, t

and

$\theta(t) = ab(t)$ and represents a time dependent function used to fit the current zero curve.

Convexity Adjustments

The Ho-Lee model eurodollar convexity adjustment is as follows

$$ConvAdj_{(HL)} = \frac{1}{2}T_1T_2\sigma^2 \quad (3)$$

and the corresponding Hull-White 1 factor adjustment is below

$$ConvAdj_{(HW1F)} = \frac{B(T_1, T_2)}{T_2 - T_1} \left(B(T_1, T_2) (1 - e^{-2aT_1}) + 2aB(0, T_1)^2 \right) \frac{\sigma^2}{4a} \quad (4)$$

where

$$B(t, T) = \left(\frac{1 - e^{-a(T-t)}}{a} \right)$$

Special Case: Mean Reversion, $a = 0$

It is easy to see that when the mean reversion parameter $a = 0$ that the Hull-White 1F model and Ho-Lee models are equivalent.¹ What remains to be seen is if the corresponding convexity corrections also agree.

Inspecting the the Hull-White convexity adjustment we can see that the we cannot substitute $a = 0$, since dividing by $a = 0$ is undefined, therefore we consider the limiting case.

$$\begin{aligned} Limit, L &= \lim_{a \rightarrow 0} (ConvAdj_{(HW1F)}) \\ &= \lim_{a \rightarrow 0} \left(\frac{B(T_1, T_2)}{T_2 - T_1} \left(B(T_1, T_2) (1 - e^{-2aT_1}) + 2aB(0, T_1)^2 \right) \frac{\sigma^2}{4a} \right) \quad (5) \end{aligned}$$

Firstly we note that

$$\begin{aligned} \lim_{a \rightarrow 0} (B(t, T)) &= \lim_{a \rightarrow 0} \left(\left(\frac{1 - e^{-a(T-t)}}{a} \right) \right) \\ &= \lim_{a \rightarrow 0} \left(\left(\frac{(T-t)e^{-a(T-t)}}{1} \right) \right) \text{ using L'Hôpital's rule} \\ &= T - t \end{aligned} \quad (6)$$

¹Substitute $a = 0$ in equation (2) and compare with equation (1).

Therefore

$$B(T_1, T_2) = T_2 - T_1 \quad (7)$$

and

$$B(0, T_1) = T_1 \quad (8)$$

Substituting the limits from equations (7) and (8) into (5) leads to

$$\begin{aligned} L &= \frac{T_2 - T_1}{T_2 - T_1} \left((T_2 - T_1)(1 - 1) + 0 \right) \frac{\sigma^2}{0} \\ &= \left((T_2 - T_1)(1 - 1) + 0 \right) \frac{\sigma^2}{0} \\ &= \frac{0}{0} \end{aligned}$$

Therefore we apply L'Hôpital's rule on equation (5) to get

$$\begin{aligned} L &= \lim_{a \rightarrow 0} \left(\frac{B(T_1, T_2)}{(T_2 - T_1)} \left(2T_1 B(T_1, T_2) e^{-2aT_1} + 2B(0, T_1)^2 \right) \frac{\sigma^2}{4} \right) \\ &= \lim_{a \rightarrow 0} \left(\frac{(T_2 - T_1)}{(T_2 - T_1)} \left(2T_1 (T_2 - T_1) e^{-2aT_1} + 2T_1^2 \right) \frac{\sigma^2}{4} \right) \text{ using limits (7) and (8)} \\ &= \frac{(T_2 - T_1)}{(T_2 - T_1)} \left(2T_1 (T_2 - T_1) + 2T_1^2 \right) \frac{\sigma^2}{4} \\ &= \left(2T_1 T_2 - 2T_1^2 + 2T_1^2 \right) \frac{\sigma^2}{4} \\ &= \frac{1}{2} T_1 T_2 \sigma^2 \end{aligned}$$

Thus we conclude that the convexity correction under the Hull-White 1 factor model is the same as that under the Ho-Lee model in the special case when the mean reversion parameter $a=0$ i.e. when the models are equivalent.

References

- Hull J** (1990a) "Pricing Interest Rate Derivative Securities", Review of Financial Studies 3(4), 573-92
- Vasicek O** (1977) "An Equilibrium Characterization of the Term Structure of Interest Rates", Journal of Financial Economics 5, 177-88